Milestone 1

Proof of Achievement Report

Project Catalyst #1200045: MPC as a Layer 2 Service to Cardano
Profila AG x Partisia Blockchain
Date: 2026-03-01

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1. Executive Summary

Milestone 1 delivers a fully functional Aiken smart contract deployed to Cardano preprod testnet. The contract creates an on-chain MPC initiation record that the off-chain relay (Milestone 2) monitors to bridge Cardano requests to Partisia Blockchain's MPC infrastructure.

All 6 acceptance criteria are met.

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2. Acceptance Criteria - Evidence Matrix

#	Criterion	Status	Evidence
1	Aiken unit tests pass	PASS	Section 4 - 6/6 tests pass
2	Contract has MPC initiation	PASS	Section 3 - validator source
3	PDF documentation	PASS	This document
4	Test results in docs	PASS	Section 4 + test-results-m1.txt
5	YouTube video of demo	PASS	youtu.be/QRj4tAt23Aw
6	Testnet execution on-chain	PASS	Section 5 - TX on CardanoScan

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3. Smart Contract - Technical Summary

3.1 Contract Details

Property	Value
Language	Aiken v1.1.21
Plutus Version	v3
Validator Hash	b8bd6946bb4f9ac805f242d86fea090191e64e1250aa8f2d54fc8c67
Script Address	addr_test1wzut662xhd8e4jq97fpdsml2pyqerejwzfg24red2n7gceczhkwaz
Source File	cardano/validators/profila_mpc.ak
Test File	cardano/validators/profila_mpc_tests.ak

3.2 Datum Type - MpcRequest

The contract uses an inline datum to record MPC initiation requests:

Field	Type	Purpose
dataset_id	ByteArray	Identifies the Profila dataset to query

query_type	ByteArray	"age_threshold" or "survey_match"
initiator_pkh	VerificationKeyHash	Public key hash of the requester
partisia_contract	ByteArray	Target PBC contract address
timestamp	Int	POSIX timestamp in milliseconds

3.3 Validation Rules

The InitiateMPC redeemer enforces 4 validation checks:

1. Initiator must sign - initiator_pkh must be in tx.extra_signatories

2. Dataset must exist - dataset_id must not be empty

3. Valid query type - must be "age_threshold" or "survey_match"

4. Positive timestamp - must be greater than 0

Each check uses Aiken's trace operator (?) so failures are logged with the specific condition that failed (e.g., "initiator_signed ? False").

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4. Unit Tests - Results

Command: cd cardano && aiken check

Result: 6/6 PASSED

#	Test Name	Expected	Result	Mem	CPU
1	test_initiate_mpc_valid	pass	PASS	37,784	12,233,900
2	test_initiate_mpc_empty_dataset	fail (empty dataset)	PASS	32,854	10,561,421
3	test_initiate_mpc_wrong_signer	fail (PKH mismatch)	PASS	33,578	10,818,873
4	test_initiate_mpc_bad_query	fail (bad query_type)	PASS	39,875	12,829,921
5	test_initiate_mpc_bad_timestamp	fail (timestamp=0)	PASS	43,796	14,301,528
6	test_valid_query_types	pass (utility fn)	PASS	7,811	1,826,212

Test Trace Output (failure cases)

Test 2: dataset_not_empty ? False

Test 3: initiator_signed ? False

Test 4: valid_query_type ? False

Test 5: timestamp_positive ? False

These traces confirm each validation rule independently rejects invalid inputs.

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5. On-Chain Deployment - Evidence

5.1 Transaction Details

Property	Value
TX Hash	50d28fd2bd263a84485b45280afd19bb4c3e20c24e9a4b52b6eac20418d53a57
Block Height	4,468,996
Slot	116,719,755
Network	Cardano Preprod
Fee	169,505 lovelace (0.169505 tADA)
Valid Contract	true

5.2 Verification Links

CardanoScan TX:

https://preprod.cardanoscan.io/transaction/50d28fd2bd263a84485b45280afd19bb4c3e20c24e9a4b52b6eac20418d53a57

Script Address:

https://preprod.cardanoscan.io/address/addr_test1wzut662xhd8e4jq97fpdsml2pyqerejwzfg24red2n7gceczhkwaz

5.3 UTxO at Script Address

The transaction created a UTxO at the script address with:

Amount: 2,000,000 lovelace (2 tADA)

Inline Datum (CBOR):


```
d8799f4f70726f66696c615f746573745f76314d6167655f7468726573686f6c64581cfc1c993261d3d3104bcdbf3f54d16c134f634360e9b40a23289432244
01b0000019cab7101a2ff
```


5.4 Decoded Inline Datum

Field	Decoded Value	Notes
dataset_id	profila_test_v1	
query_type	age_threshold	
initiator_pkh	fc1c993261d3d3...28943224	(28-byte key hash)
partisia_contract	(empty)	(to be filled in M2)
timestamp	2026-03-01T22:07:15Z	POSIX ms

5.5 Independent Verification

Anyone can verify this deployment via Blockfrost API:

```
curl -H "project_id: <YOUR_KEY>" \

"https://cardano-preprod.blockfrost.io/api/v0/txs/50d28fd2bd263a84485b45280afd19bb4c3e20c24e9a4b52b6eac20418d53a57/utxos"
```

Or via CardanoScan (no API key needed) using the links in Section 5.2.

6. Repository

GitHub: https://github.com/Profila/Aiken-Partisia_MPC

Contract source: cardano/validators/profila_mpc.ak

Test source: cardano/validators/profila_mpc_tests.ak

Deploy script: cardano/scripts/deploy-m1.ts

Test results: cardano/tests/test-results-m1.txt

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7. How This Enables Milestones 2-4

The deployed UTxO at the script address serves as the on-chain trigger for the entire MPC pipeline:

```
M1 (this milestone)       M2 (next)               M3 + M4
UTxO with MpcRequest  ->  relay.ts detects it  ->  Secrets submitted to PBC
inline datum              calls PBC REST API       MPC computation runs
                                                   Result displayed
```

The relay script (Milestone 2) polls this address via Blockfrost. When it finds a UTxO with a valid MpcRequest datum, it bridges the request to the Partisia Blockchain contract - triggering the privacy-preserving computation.

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8. Video Demo

M1 Demo: https://youtu.be/QRj4tAt23Aw

Combined M1-M4 Demo: https://youtu.be/jwwfdTSUS2w

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